

**Project Overview:**

The term project is an opportunity to investigate and analyze a mechanics problem of your choice using ABAQUS®. The project topic can be anything of your choice, it can be based on your ongoing research, or it can be a topic that interests you. The primary purpose of the project is to evaluate your model building skills using ABAQUS®.

This involves:

- 1) Introducing the problem and its importance
- 2) describing the model development
- 3) Analysing and interpreting the results

**Project Report Weight:** 30%

**Formatting:**

- Cover Page: (Project Title, Name and ID)
- Page size: Letter (8.5x11”) with 1” margin
- Font Size: 11
- Line Spacing: 1.5
- Include descriptive captions for tables and figures
- If you use any equations please number them
- Make sure you introduce any symbols
- You can include an optional nomenclature in the appendix if there are too many symbols
- Include a page number: Introduction (P1) to Conclusion (Pn)
- Page Limit: 15
  - Please do not exceed the page limit
  - This is an upper limit, you do not need to have 15 pages. Ideally, the report should be more concise
  - The limit does not include (Cover page, Abstract, References and Appendix)
- Properly cite references using in text citations and include them in the references section (You can use any citation style but please be consistent)
- Use numbered headers and sub-headers for each report section
- If you plan to use acronyms, make sure you introduce the acronym the first time you use it

**Report Structure:**

The report should follow the structure of a scientific paper: Abstract → Introduction → Methods → Results → Discussion → Conclusion.

- Abstract
  - On a separate page
  - 500 Word Abstract
  - 2nd Page in the report (Not numbered)
  - Should contain: ▪ Brief introduction to the problem, objective, highlight the methods, main findings and concluding remarks
- Introduction
  - Page numbering starts from the introduction.
  - Consists of three subsections
    - Background
    - Motivation
    - Objectives and Project Scope
  - Please ensure that proper references are used to support the introduction and to highlight previous work in the literature
- Methods
  - Consists of the following sections
    - Problem's Domain
      - Main component/components of the system
      - Simplifying assumptions if applicable
      - If there is more than one component, please mention how they interact with each other: for example contact, cohesion, tied (no relative motion)
      - Support this section with a figure/schematic showing the system components and major (characteristic) dimensions
    - Material Properties
      - Describe (preferably in a tabular manner) the material model and properties of each component.
      - Briefly justify the choice of the material model
    - Interactions
      - This section is only applicable if you have more than one component interacting with each other
      - You will need to describe in more details the interactions between the different components for example:
        - Tied surfaces: Where are the tied regions and which one is the primary surface and which one is the secondary surface

- Support this section with figures/schematic showing interacting surfaces
  - Contact: Which surfaces are in contact? What type of contact module?
- Analysis Procedure
  - Include the type of analysis: Static, Eigenvalue, Dynamic (Implicit/Explicit)
  - If your simulation is dynamic, you may need to justify/check the choice of the time step.
- Meshing
  - Specify the chosen element
    - 1D/2D/3D
    - Type: Solid Quadrilateral, Solid Tetrahedral, Solid Hexahedral, Shell, Beam and etc.
    - Order: Linear/Quadratic
    - Integration: Reduced or Full
  - Justify the element choice
  - Describe how you plan to conduct mesh independence
  - Support this section with a figure showing the mesh of the analyzed part/parts
- Loading and Boundary Conditions
  - State the boundary and loading conditions
  - Support this with a schematic/figure showing these conditions
- Validation Plan
  - Conclude this section by briefly stating how are you planning to validate your model for example:
    - Comparison with the literature
    - Comparison with an experiment
    - Analytical Validation
    - Comparison to a Standard/Code
- Results
  - Divide this section in a logical manner that follows your objective/objectives.
  - Include the results of mesh independence before you present any results. If there is not enough space, you can write a brief paragraph describing the results of mesh independence and include tables/figures in the appendix
  - Present the results (using tables and figures) and describe them qualitatively and quantitatively.
  - You can include any comparisons with the literature or experiment but leave the detailed discussion for the discussion section
- Discussion
  - The discussion section will follow the same manner and order of your objectives and results section
  - It should not include new results/findings but rather it focuses on analyzing the reported results

- In this section you need to give more detailed qualitative and/or quantitative analysis of your results
    - Are the results making sense quantitatively/qualitatively?
    - Are they close to the experimental and or literature values?
    - If there are discrepancies, what might be causing it?
  - The discussion should be supported by references
  - Conclude this section with recommendation for future work if applicable
- Conclusion
  - Conclude this section with recommendation for future work if applicable
- References
- Appendix (If needed)
  - You can include additional figures/tables that did not fit in the report
  - You can use this section to show sample calculations if you did analytical calculations or was using an engineering code/standard
- AI Usage Disclosure Form
  - As discussed at the beginning of the semester, the use of AI and LLMs is only permitted for aiding interpretation, identifying references or search terms, and polishing or proofreading your own writing. The generation of entire sections or substantial content using AI is not permitted.
  - For more questions about proper AI usage, feel free to reach out with question.
  - Please complete, sign, and attach the AI usage disclosure form at the end of your report
  - The form will be provided separately on Canvas

|                                   | Unacceptable (0-25%)                                                                              | Developing (25-60%)                                                                             | Acceptable (60-85%)                                                                                                                      | Exemplary (85-100)                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Formatting (20 %)</b>          |                                                                                                   |                                                                                                 |                                                                                                                                          |                                                                                                                            |
|                                   | No adherence to any formatting guidelines                                                         | Some Adherence to formatting guidelines                                                         | Adherence to formatting guidelines with minor discrepancies                                                                              | Adhered to all formatting guidelines                                                                                       |
| <b>Language and Clarity (20%)</b> |                                                                                                   |                                                                                                 |                                                                                                                                          |                                                                                                                            |
|                                   | Major issues throughout the entire report<br><br>Majorly affected my understanding of the project | Major issues throughout the report that did not affect my overall understanding of the project. | Overall, language is clear but there are a few major typos/mistakes<br><br>Proper use of visual aids and captions but it can be improved | Language is clear and concise<br><br>Very few typos/grammatical mistakes<br><br>Proper use of visual aids (figures/tables) |
| <b>Content (60%)</b>              |                                                                                                   |                                                                                                 |                                                                                                                                          |                                                                                                                            |
| Abstract                          | No Abstract                                                                                       | Met some of the outlined requirements but highly exceeded the word limit                        | Met most of the requirements and slightly exceeded the word limit                                                                        | Met all of the requirements including the word limit                                                                       |
| Introduction                      | Missing introduction or one of its main sections                                                  | Met some of the outlined requirements                                                           | Met most of the requirements                                                                                                             | Met all of the requirements                                                                                                |
| Methods                           | No methods or missing major sections of it                                                        | Met some of the outlined requirements                                                           | Met most of the requirements                                                                                                             | Met all of the requirements                                                                                                |
| Results                           | No results or missing major sections of it                                                        | Met some of the outlined requirements                                                           | Met most of the requirements                                                                                                             | Met all of the requirements                                                                                                |
| Discussion                        | No discussion or missing major sections of it                                                     | Met some of the outlined requirements                                                           | Met most of the requirements                                                                                                             | Met all of the requirements                                                                                                |

|            |               |                                       |                              |                             |
|------------|---------------|---------------------------------------|------------------------------|-----------------------------|
| Conclusion | No conclusion | Met some of the outlined requirements | Met most of the requirements | Met all of the requirements |
|------------|---------------|---------------------------------------|------------------------------|-----------------------------|

Note: Please note that every project is unique, if you think one of the grading criteria does not strictly apply to you feel free to reach out